

Review

Interventions to improve attitudes toward older people in undergraduate health and social sciences students. A systematic review and meta-analysis

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ABSTRACT

Objective: To determine which interventions are the most effective in improving attitudes toward older persons in undergraduate health and social sciences students.

Design: Systematic review and meta-analysis.

Data sources: A literature search was made in PubMed, EBSCO and SCOPUS and additional records were identified by manual searching. The selection criteria were studies that evaluated an intervention designed to improve positive attitudes; studies in undergraduate health and social sciences students; and studies using direct instruments to measure attitudes.

Review methods: Two independent reviewers extracted study-level data from records using a common data collection spreadsheet. Two reviewers also assessed study quality by using the Cochrane Risk of Bias Tool. The primary outcome was the effect of an intervention upon attitudes toward older people, while the secondary outcome was the effect upon knowledge about aging and older people in those studies that had previously assessed subject attitude. The meta-analysis was carried out based on mixed statistical models.

Results: The search identified 53 eligible studies published during 1982–2020, comprising 35 pre-post studies and 18 randomized clinical trials, of which 14 were included in the meta-analysis. A strongly significant effect upon attitudes was observed for empathy-based interventions (differences of standardized mean differences (d_{SMD}) = 1.26; 95%CI: 0.04–2.48; p = 0.04), knowledge + empathy-based interventions (d_{SMD} = 0.22; 95%CI: 0.05–0.39; p = 0.01), and knowledge + clinical skills-based interventions (d_{SMD} = 0.22; 95%CI: 0.01–0.43; p = 0.04). The overall effect was d_{SMD} = 0.50; 95%CI: 0.01–0.43; p = 0.004. In addition, a positive effect in terms of increased knowledge about the older was observed after knowledge + empathy-based and knowledge + clinical skills-based interventions (d_{SMD} = 0.24; 95%CI: 0.07–0.40; p = 0.005).

Conclusions: Interventions to improve attitudes in health and social sciences students are effective, with those based on empathy having the greatest impact. Improving attitudes among future professionals could improve the management and quality of care of older people. More rigorous and better designed studies are recommended to determine the effect of the interventions.

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Twittable abstract: Ageism is present in healthcare settings. Empathy-based interventions are the most effective strategies in health and social sciences students.

1. Background

Aging is a natural and irreversible process that involves

physiological, psychological (both emotional and cognitive) and social changes, and may take a course that can be defined as successful, typical or pathological. In pathological aging, the mentioned changes tend to

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predispose to a higher incidence of chronic disease conditions, which in turn generate a greater need for healthcare (Dziechciaż and Filip, 2014).

The growth of the population over 60 years of age has been a fact for years (World Health Organization, 2018). However, the COVID-19 pandemic has further highlighted the large number of older people with a greater susceptibility to vulnerability, evidenced by the need for example to provide them with schedules to go shopping, or to prioritize them in the corresponding vaccination strategies (Monahan et al., 2020). On the other hand, there are also obvious negative aspects such as underrepresentation and undervaluing in the current public discourse on the pandemic; problems with documenting deaths among older adults; a lack of preparation of long-term care facilities; the existence of patronizing “protective” policies; and the initial public perception that the viral infection was really a problem of older adults (Fraser et al., 2020). This discriminatory attitude toward the older with respect to the rest of the population is referred to as “ageism”, and is found both in the clinical setting and in functional, psychological and social aspects (Monahan et al., 2020; Van Wicklin, 2020). Ageism is defined as “negative” or positive stereotypes, prejudice and/or discrimination against (or to the advantage of) older people on the basis of their chronological age or on the basis of a perception of them as being “old” or “elderly”. Ageism can be implicit or explicit, and can be expressed on a micro-, meso- or macro-level (Iversen et al., 2009).

The World Health Organization (WHO) launched a campaign against ageism in 2018 (Officer and de la Fuente-Núñez, 2018). The condition of being an older patient is different from the condition of being an older person, and this leads us to postulate that there are stereotypes, prejudices and discriminatory practices specifically related to the condition of being an older patient - with direct consequences for their treatment and care in the healthcare setting. Reducing all the concepts encompassed by ageism among healthcare professionals is a pressing need (São José et al., 2019).

Stereotypes of older adults, and the assumption that the older are vulnerable and dependent, affect both the society that treats older adults and healthcare professionals. Older adults should be seen as worthy of respect, and ageism can affect the physical, emotional and social quality of life of older individuals (Van Wicklin, 2020). Numerous studies have been conducted to evaluate attitudes toward the care of the older among healthcare professionals in different disciplines, including medicine, social work, pharmacy, nursing, etc. Although there may be certain differences among these disciplines, the literature proposes training as one of the solutions to reduce anti-aging bias (Scott et al., 2019).

Healthcare professionals such as physicians and nurses are the professionals that spend most time with patients (Butler et al., 2018; Westbrook et al., 2011). If negative attitudes toward this population interfere with the care of the older (Van Wicklin, 2020), interventions targeted to undergraduate health and social care students become necessary to address the problem (Ross et al., 2018a, 2018b; Samra et al., 2013, 2017). In relation to this situation, the literature evidences a dual perspective: health and social care students a priori show a good disposition to work with older adults, though on the other hand a certain degree of ageism is present in the care of the older, such as the tendency to speak slower and louder in their presence. Several reviews have examined different interventions in undergraduate students, but the results are inconclusive due to difficulties in comparing outcomes and in the design of the studies (Eymard and Douglas, 2012; Ross et al., 2018a, 2018b; Samra et al., 2013, 2017; Seah et al., 2019; Tullo et al., 2010; Wyman et al., 2018). A meta-analysis of interventions in any population has concluded that they are effective in reference to attitudes, knowledge and comfort, but do not reduce anxiety and/or reluctance to working with older people. Moreover, the strongest effects after the interventions were observed in women, adolescents and young people (Burnes et al., 2019).

To date, no systematic review and meta-analysis has analyzed the types of interventions that are most effective in undergraduate health and social sciences students, were female and young individuals are

prevalent (Salvi et al., 2020). The present systematic review was carried out to determine which interventions are the most effective in improving attitudes toward older persons among undergraduate health and social sciences students.

2. Materials and methods

The present study was designed and carried out according to the guidelines of the Cochrane Handbook for Systematic Reviews of Interventions. The details of the protocol for this systematic review were registered on PROSPERO ID: CRD42021220677.

2.1. Literature search

The PICO format (population/intervention/comparison/outcomes) was used to define the research question (Supplementary file 1, Table S1.1).

A literature search was carried out in multiple electronic databases (Medline, Scopus and EBSCO) of all entries up until 17 November 2020, with no publication age limitations. Each database was searched using the terms shown in Supplementary file 1 (Table S1.1), as a single search term or in combination using Medical Subjects Heading (MeSH) terms with the Boolean operators AND/OR.

The reference lists of all relevant articles were manually cross-referenced in order to identify possible additional articles. The search for unpublished studies included an electronic search of trial records: current controlled trials (<http://www.controlled-trials.com>), the National Institute of Clinical Health Databases (<https://clinicaltrials.gov>), the Universal Index of Doctoral Dissertations in Progress, Mednar, review of the gray literature and Google Search.

2.2. Inclusion and exclusion criteria

In order to answer our research questions, we applied the following inclusion criteria to each of the considered manuscripts: (1) articles acknowledged as peer-reviewed original papers and including experimental studies (randomized and controlled trials) and pre-post intervention studies; (2) full-text articles published in English or Spanish; (3) studies assessing attitudes toward older persons after the implementation of any type of intervention; (4) articles providing a description in the methods section of how attitudes were evaluated; and (5) studies in which the type of intervention was specifically described. Studies in animals, case reports, qualitative studies, letters to the editor, abstracts from conferences, books and doctoral theses were excluded. In addition, we excluded those studies that analyzed attitudes with an indirect measurement instrument (f.i. to use a scale that assesses knowledge or work preference to assess attitudes).

2.3. Data collection and analysis

Details on the manuscripts identified in the literature search were uploaded into a web-based system that was used to manage the screening process and remove any duplicate citations. To determine which studies would be included, four members of the review team (L-L-H and E-C-R, M.B-N and FM-M-A) independently screened the titles and abstracts of these articles and, based on our inclusion and exclusion criteria, the corresponding full texts were retrieved. Two reviewers (L-L-H and E-C-R) independently extracted the following data from each article: date of publication and type of study, the country in which the study was conducted, number and characteristics of the students included (type of student, age and sex), description and type of intervention, attitude assessment scale, validity and reliability of the instrument in the sample, and the main (attitude) and secondary outcome (knowledge) of the study. In order to facilitate comparison of the results, the authors categorized the interventions as follows: knowledge-based, empathy-based, clinical skills-based, or a combination of these.

For the main outcome (attitude), in both pre-post studies and in RCTs we collected the pre-and post-intervention data reported by the authors of the studies, with mean and standard deviations or percentages, mean differences and significance. In those cases, in which the authors failed to report the mean differences or percentages, but sufficient information was available, we personally calculated these data. To facilitate the comparison of results in the pre-post intervention studies, the authors calculated the percentage change between 0 and 100. To do this, the ranges of scores for each scale were determined and the percentage change was calculated in each case. For example, the Aging semantic differential (ASD) scale scores between 23 and 115, so the change produced was calculated on the basis of 92 points. Likewise, the Questionnaire on negative stereotypes about aging scale (CENVE) scores between 15 and 60, so the change produced was calculated on the basis of 45 points. In this way, the changes produced could be compared despite the large number of scales used by the different authors. In addition, there were both positive (the higher the score, the better the attitude) and negative scales (the higher the score, the poorer the attitude); accordingly, in the case of negative rating scales, we inverted the scores (Supplementary file 2).

In the comparison of results in the meta-analysis, we entered those studies that reported means and standard deviations in both the intervention and the control groups. In those cases, in which the scales used were negative (i.e., the higher the score, the poorer the attitude), the data entered in the meta-analysis were inverted according to the groups (CG x IG) to allow comparison with the rest of the studies that used positive scales.

The secondary outcome (increase in knowledge) was analyzed only

in the RCTs, since the pre-post studies lacked a control group, and there was greater bias in the reliability of the results (Samra et al., 2013).

Any disagreement between the reviewers regarding the papers and data extracted from them was resolved by consulting other author (P-P-R).

2.4. Assessment of risk of bias in the selected studies

Two reviewers independently rated the quality of the studies using the Cochrane protocol for assessing risk of bias (RoB2). The RoB2 tool comprises 7 domains: random sequence generation; allocation concealment; blinding of participants and personnel; blinding of outcome assessment; incomplete outcome data; selective reporting; and other sources of bias. Each evaluation was classified by two independent authors as corresponding to high risk of bias, low risk of bias, or unclear bias.

The judgments were made independently by two researchers (FM-M-A and M.B-N), with any discrepancies being resolved through discussion with the corresponding author (P-P-R).

3. Results

3.1. Characteristics of the studies

The literature search retrieved a total of 593 articles and 37 additional records through other sources. After the study selection process, 53 articles were included in the analysis. The full study selection process is presented in Fig. 1 as per the recommended preferred reporting items

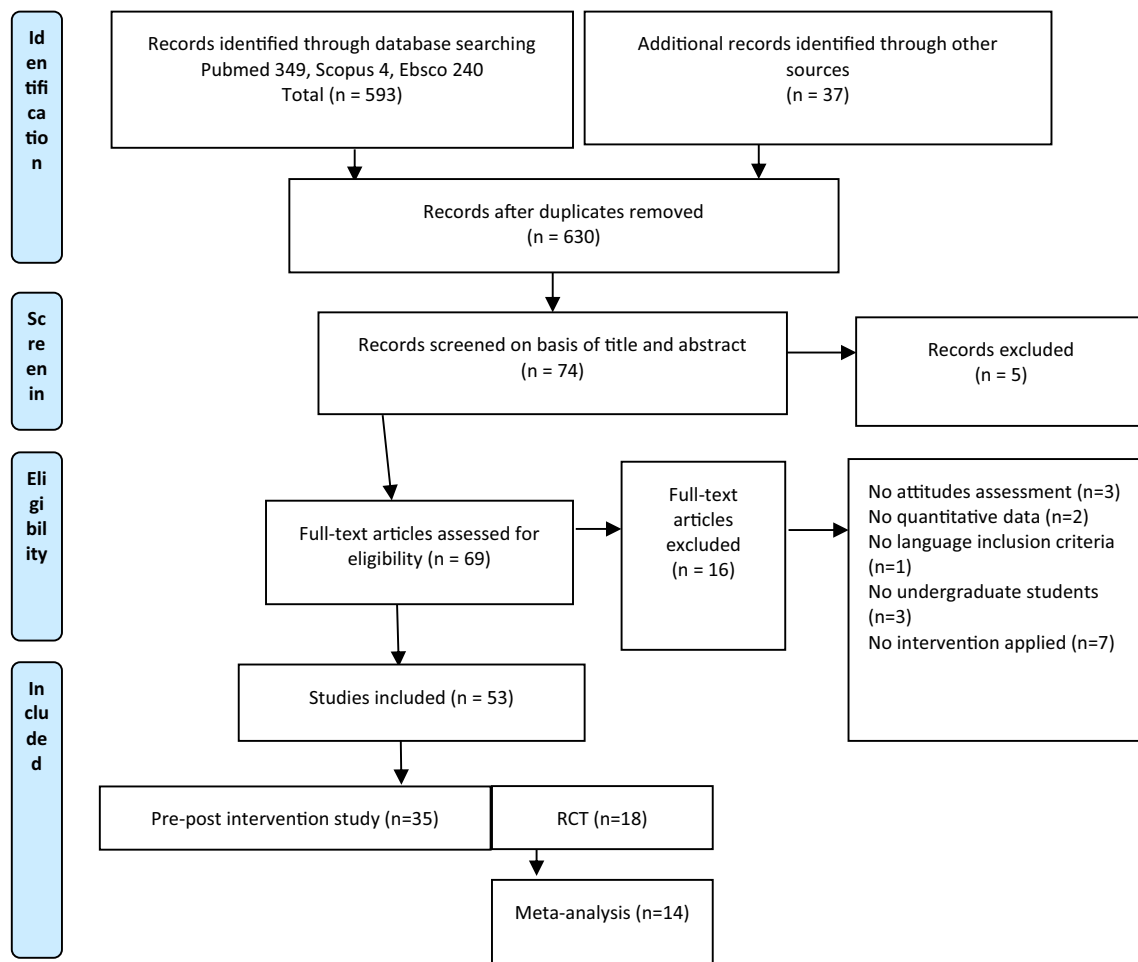


Fig. 1. Preferred reporting items for systematic reviews and meta-analyses (PRISMA) flow diagram.

for systematic reviews and meta-analyses (PRISMA) statement [29].

Details of the study characteristics are presented in Tables 1 and 2. The included studies were published between 1982 and 2020. Eighteen studies were RCTs, while 35 were pre-post intervention studies comprising one to four groups.

Most of the studies carried out their intervention in medical students (Alford et al., 2001; Bernard et al., 2003; Deary et al., 1993; Denton et al., 2009; Diachun et al., 2010; Duke et al., 2009; Eskildsen and Flacker, 2009; Fields et al., 1992; Goeldlin et al., 2014; Gonzales et al., 2010a; Hughes et al., 2008; Intrieri et al., 1993; Jeste et al., 2018; Lee et al., 2005; Lindberg and Sullivan, 1996; Lorraine et al., 1998; Lu et al., 2010; Pacala et al., 1995; Robinson and Rosher, 2001; Shue et al., 2005; Stewart et al., 2007; Van De Pol et al., 2014; Warren et al., 1983; Westmoreland et al., 2009; Wilkinson et al., 2002; Wilson and Glamser, 1982); one in medical residents in training (Baum and Nelson, 2007), and one in paramedical students (Ross and Williams, 2015). Six studies included health and social sciences students (Andreoletti and Howard, 2018; Carmel et al., 1992; Dorfman et al., 2003; Even-Zohar and Werner, 2020; Henry et al., 2007; Ng et al., 2020), 12 included nursing students (Can Gür and Yilmaz, 2020; Chen et al., 2015; Demirtas and Basak, 2020; Denton et al., 2009; Gholamzadeh et al., 2018; Haight et al., 1994; Hollar et al., 2011; Hwang et al., 2013; Ryan et al., 2007; Sarabia-Cobo and Castanedo Pfeiffer, 2015; Sari et al., 2020; Walsh et al., 2008; Williams et al., 2007), one study involved physiotherapy students (Hobbs et al., 2006), four included health sciences students (Heliker et al., 1993; Henry et al., 2011; Leung et al., 2012; Silver et al., 2016) and one also included psychology students (Ragan and Bowen, 2001). Lastly, one study included students in dietetics and nutrition (Lee et al., 2008), and one included health sciences postgraduate students (Fitzgerald et al., 2006).

Most of the studies analyzed populations between 20 and 25 years of age, and with a higher percentage of female students particularly in nursing and health sciences students, covering nursing, medical and/or dietetics or physiotherapy disciplines.

With regard to the scales used to assess the main outcome “attitude”, the different authors employed a range of validated instruments. The Loyola Generativity Scale (LGS) (Andreoletti and Howard, 2018), the Relating to Older People (ROPE) (Cherry and Palmore, 2008), The Jefferson Scale of Empathy (JSE) (Hojat et al., 2001), the Positive and Negative Ageism Scale (PNAS) (Yurtas and Sarikoca, 2018), the Reaction to Aging Questionnaire (RAQ) (Gething, 1994), the Expectations Regarding Aging (ERA) (Sarkisian et al., 2002), Fraboni's Scale of Ageism (FSA) (Fraboni et al., 1990), Kogan's Scale of Attitudes Toward Older People (KAOP) (Kogan, 1961), the Aging Semantic Differential (ASD) (Rosencranz and McNevin, 1969), the Refined-Aging Semantic Differential (R-ASD) (Gonzales et al., 2010b) and Polizzi's Refined Version of the Aging Semantic Differential (ASD-Polizzi) (Polizzi, 2003), the Carolina Opinions on Care of Older Adults (COCO) (Hollar et al., 2011), the UCLA-GAS (Geriatric Attitudes Scale) (Reuben et al., 1995), the Questionnaire of Negative Stereotypes about Aging (CENVE) (Blanca-Mena et al., 2005), the Maxwell-Sullivan Attitudes Survey (MSAS) (Maxwell and Sullivan, 1980), Kafer's Aging Opinion Survey (Kafer et al., 1980), and the Wall-Oyer Aging Inventory (WOAI). Some authors used a pilot instrument developed for the measurement of attitudes (Alford et al., 2001; Carmel et al., 1992; Deary et al., 1993), and others used modified versions of validated instruments (Baum and Nelson, 2007; Stewart et al., 2007; Warren et al., 1983) (Supplementary file 2).

With regard to the secondary outcome “knowledge”, most of the authors used the Facts on Aging Quiz (FAQ) developed by Palmore (Palmore, 1977), in addition to the UCLA Geriatric Knowledge Test (GKT) (Lee et al., 2004).

The interventions were highly varied and involved different durations ranging from a single one-hour session to extra training lasting several weeks, or curricular programs covering several years. Most of the studies used interventions based on knowledge or a combination of

knowledge and empathy (Tables 2 and 3).

3.2. Pre-post intervention studies

Comparison of the results of the different studies proved difficult not only because of the utilization of different evaluation instruments but also due to the fact that different authors made different uses of one same instrument. For example, validation of the KAOP scale contemplates scores ranging from 34 to 204 (Kogan, 1961), but some authors scored over a different range from 24 to 170 (Ng et al., 2020; Ryan et al., 2007), while Hilt et al. employed a revised version of the KAOP (Hilt and Lipschultz, 1999) with an inverted score from 22 to 132, i.e., a lesser score indicated a better attitude. Despite this, we were able to establish comparisons after calculating percentage improvement according to the post-intervention difference obtained, based on the scoring range of the scale used (Supplementary file 2).

Regarding the pre-post studies, most of the articles reported significant improvements in attitude (Table 3). Only the study published by Hobbs (Hobbs et al., 2006) recorded a decrease in positive attitude, though statistical significance was not reached. The greatest increases in positive attitude corresponded to the knowledge-based intervention studies (Haight et al., 1994; Ragan and Bowen, 2001; Sarabia-Cobo and Castanedo Pfeiffer, 2015), with a heterogeneous duration ranging from 2 to 4 weeks (Ragan and Bowen, 2001) to three years of university education (Haight et al., 1994). Positive increments were also recorded in the empathy-based interventions (Chen et al., 2015; Henry et al., 2007; Lorraine et al., 1998; Sari et al., 2020; Walsh et al., 2008), and in those based on knowledge + empathy (Alford et al., 2001; Baum and Nelson, 2007; Hwang et al., 2013; Silver et al., 2016). The studies that reported comparatively less positive outcomes were those based on knowledge + clinical skills (Deary et al., 1993; Fields et al., 1992; Goeldlin et al., 2014; Hobbs et al., 2006; Hughes et al., 2008; Lee et al., 2005).

3.3. Randomized clinical trials

Of the 18 RCTs included in our study, four (Bernard et al., 2003; Denton et al., 2009; Heliker et al., 1993; Pacala et al., 1995) lacked the required post-intervention data or standard deviations, and therefore could not be included in the meta-analysis. On the other hand, in three articles (Even-Zohar and Werner, 2020; Intrieri et al., 1993; Lu et al., 2010) we exchanged the data of the tables (CG in place of IG) in order to be able to compare them with the rest of the studies, since the scales used by the latter were of a negative design (higher scores indicating poorer attitudes) (Fig. 2).

In relation to the improvement of attitude toward older people, the pooled data indicated strong evidence of a difference between intervention and control (standardized mean difference = 0.50, 95%CI: 0.16–0.84; $p = 0.004$ [heterogeneity: $\text{Tau}^2 = 0.37$; $\text{Chi}^2 = 132.38$, $\text{df} = 14$; $p < 0.00001$; $I^2 = 90\%$]; $k = 14$; $n = 1541$). The interventions based on empathy showed the greatest increases in positive attitude, though heterogeneity was found to be greater (standardized mean difference = 1.26, 95%CI: 0.04–2.48; $p = 0.04$ [heterogeneity: $\text{Tau}^2 = 1.49$; $\text{Chi}^2 = 87.52$, $\text{df} = 4$; $p < 0.00001$; $I^2 = 97\%$]) (Fig. 2).

3.4. Risk of bias

In general, the studies included in this meta-analysis presented low risk of bias according to the Cochrane risk of bias tool. The majority of studies with high or unclear risk of bias were based on quasi-experimental designs, and therefore demonstrated shortcomings related to randomization, allocation concealment, and blinding of participants, personnel and outcome assessors (Fig. S1.2; Supplementary file 1).

Table 1

Characteristics of the pre-post intervention studies.

Study (author and year)	Country	Study design	Population type	Population	Population, mean age (SD-range)/percentage of groups of age	Population % women	Intervention description	Duration
Alford et al., 2001	USA	P-P I One G	Medical students	$n = 204$	na	52.45	Geri Track Program: 1-hour didactic presentation by a geriatrician, a visit to an assigned older volunteer and a response to reflection questions	1 h
Andreoletti and Howard, 2018	USA	P-P I One G	Social and health sciences	$n = 50$	na	na	Two focus discussion groups students and older adults of 20–25 min and a focus discussion group with all participants	60–75 min
Baum and Nelson, 2007	USA	P-P I One G	Medical residents	$n = 67$	na	na	12 teaching session half day per month and 1 resident (OIA) every 2 months	1 year
Carmel et al., 1992	Israel	P-P I Three G	Social and health sciences	$n = 71$ (GM $n = 35$, GN $n = 24$, GSW $n = 12$)	GM: 23,46 (2,09), GN: 34 (8,05) GSW: 26,18 (4,58)	GM 34.3. GN 99 GSW 90.9	GM (group of medical students): 25-h course, including lectures, discussions and interviews of older people (OCA; OIA; OHA: older community/ institutionalized/hospitalized adults). GN (group of nursing students) and GSW (group of social worker students): 24 h lecture course in gerontology	GM and GN 25 h and GSW 24 h
Chen et al., 2015	USA	P-P I One G	Nursing students	$n = 30$	22.46 (1.47)	50	Simulation	One course
Deary et al., 1993	UK	P-P I One G	Medical students	$n = 64$	na	50	Training block in geriatric medicine	4 weeks
Duke et al., 2009	USA	P-P I One G	Medical students	$n = 55$	23.7	54.28	Geriatric Mentoring Program. The student visited the same older 4 times during the academic year. They discussed about topics of old age that were provided in a notebook. After each visit the student wrote a text with reflections on assigned topics.	4 sessions of 45 min during academic year
Eskildsen and Flacker, 2009	USA	P-P I One G	Medical students	$n = 129$	ranges 20–24	49.2	5 sessions on topics ranging from biological to social topics related to aging + interview with OA advanced aged each day	5 days
Fields et al., 1992	USA	P-P I One G	Medical students	$n = 127$	25 (2)	40	Pre-clinical course + clinical rotation with older patients	4 weeks
Goeldlin et al., 2014	Switzerland	P-P I Two G	Medical students	$n = 154$	22.9 SD 2.1	61.6	Geriatric Clinical Skills Training: 4 × 2.5 h teaching sessions + textbook to self-prepare	10 h
Gonzales et al., 2010a	USA	P-P I Two G	Medical Students	$n = 154$ (IG $n = 86$; CG = 68)	IC 24 (1.9), CG 24 (2.9)	IC 67 CG 55	Vital Visionaries (CI) 2-hour sessions at local with OCA and after focus groups in art museums to create and discuss art	6 months to 1 year
Haight et al., 1994	USA	P-P I One G	Nursing students	$n = 57$	20–40 age range	91.22	Nursing degree: year 1 subject: care of the healthy elderly; year 2 subject: care of the ill elderly; year 3 subject: care of the critically ill elderly	3 years
Henry et al., 2007	USA	P-P I One G	Social and health sciences	$n = 156$	81% < 25 years	84	Aging game (simulation [students assuming the roles of elderly adults with loss simulation])	3 h along 5 months
Hobbs et al., 2006	Australia	P-P I One G	Physiotherapy students	$n = 165$	21,8	70	6 clinical placements (1 × 4 wk. in 2nd year; 1x5wk in 3rd Year and 4 × 5 wk. in 4th year)	29 weeks
Hughes et al., 2008	UK	P-P I One G	Medical students	$n = 70$	100% < 25 years	54.3	GM (Geriatric Medicine) teaching program and clinical	8-day
Hwang et al., 2013	China	P-P I Three G	Nursing students	$n = 126$	17.95	na	6 h pre-service training, 12 h supervised service, 2 h interaction with assisted living facilities, nursing homes) and veteran's homes	16 months

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Table 1 (continued)

Study (author and year)	Country	Study design	Population type	Population	Population, mean age (SD-range)/percentage of groups of age	Population % women	Intervention description	Duration
Jeste et al., 2018	USA	P-P I One G	Medical students	n = 149	18 or more	55.7	MSTAR and M-Stream (didactic, clinical and research sessions on geriatrics content)	8–12 weeks
Lee et al., 2005	USA	P-P I Two G	Medical students	n = 165 (GF n = 38, GR n = 127)	na	na	Fellowship geriatrics training versus residents	1 year
Lorraine et al., 1998	USA	P-P I One G	Medical students	n = 100	na	na	Simulation in aging (older with Parkinson, advance diabetes, stroke, rheumatoid arthritis, incontinence and glaucoma)	2 weeks
Ng et al., 2020	Singapore	P-P I One G	Social and health sciences students	n = 181	21 (18–41)	68	TriGen involves participants across three generations: older patients, healthcare undergraduate students and secondary school students	6 months
Ragan and Bowen, 2001	USA	P-P I Three G	Psychology students	n = 99 (InfG = 33, CLG n = 33; RIG = 33)	20 (2.78)	63	Information Group (InfG): videotape about aging myths. Information and campus life group (CLG): video +2 discussion sessions about campus life 25 min each one, avoiding videotape content, and reinforcement information group (RIG): videotape + 2 discussion session 25 min each about videotape content and capacity for attitudinal change	Between 2 and 4 weeks depending on group
Robinson and Rosher, 2001	USA	P-P I One G	Medical students	n = 41	na	na	“Half-full Aging Simulation Experience”. Simulation in reduction of 4 types of aging skills and tricks to facilitate daily life for the older	1 course
Ross and Williams, 2015	Australia	P-P I One G	Paramedical students	n = 11	Median 20	63.6	6 × 10-minute conversations with seniors on topics of interest to them	6 × 10 min, survey after 5 weeks
Ryan et al., 2007	Ireland	P-P I One G	Nursing students	n = 94	80% ≥ 25 years	97	Theoretical 2 h session +10 visits at home OCA	1 year
Sarabia-Cobo and Castanedo Pfeiffer, 2015	Spain	P-P I One G (three cohorts)	Nursing students	n = 222	na	87.3	Geriatrics subjects in Nursing Curriculum	1 academic year
Sari et al., 2020	Turkey	P-P I One G	Nursing students	n = 58	94.8% between 19 and 21	87.9	Geriatric Medication Game	3 h
Silver et al., 2016	Canada	P-P I One G	Health sciences students	n = 51	21 (1.5)	90.2	Semester course geriatrics with a humanistic approach	13 weeks
Stewart et al., 2007	USA	P-P I Four G	Medical students	G1 n = 53, G2 = 55, G3 = 65 G4 = 68	na	na	New medical school curriculum, 2 groups received partial treatment over 2 years. G1 and G2 received half of the new geriatrics curriculum. G3 and G4 received the full new geriatrics curriculum	4 years
Van De Pol et al., 2014	Netherlands	P-P I One G	Medical students	n = 53 (GG n = 29 and GNS n = 24)	GG 23 (1.5), GNS 24 (1)	GG 51.7, GNS 33	GG (Group Geriatrics) GeriatricX (game) + education intervention versus GNS (Group in Neurosciences) education in neuroscience	4 weeks
Walsh et al., 2008	USA	P-P I One G	Nursing students	n = 22 (GCB n = 11, GFV n = 11)	na	na	Two different interventions: GCB: Group of creative bonding intervention and GFV: Group of friendly visits	4 weeks
Warren et al., 1983	USA	P-P I One G	Medical students	n = 71	na	na	Three major learning activities: an introduction to geriatrics, geriatric field placements, and lecture and discussion sessions	25 h during 6 weeks
Westmoreland et al., 2009	USA	P-P I One G	Medical students	n = 237 (GPre n = 119, GPost = 118)	na	na	Session of 90 min = 5 min writing + 5 discussion+75 min dialogue with OA + 5 min reflection only for GPost	Over a year
Wilkinson et al., 2002	New Zealand	P-P I One G	Medical students	n = 186 (GOCA n = 41, GOIA n = 40, GYP n = 105)	na	na	Early contact week: G1 interview in pairs with OCA; G2 with OIA and G3 young people	One week

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Table 1 (continued)

Study (author and year)	Country	Study design	Population type	Population	Population, mean age (SD-range)/percentage of groups of age	Population % women	Intervention description	Duration
Williams et al., 2007	Canada	P-P I One G	Nursing students	n = 38	ranges 20–29	100	CBL (context-base program): at least 6 weeks per year in different areas: real nursing situation, working with healthy older adults, with older residents in continuing care facilities, and acute care	At least 24 weeks
Wilson and Glamser, 1982	USA	P-P I One G	Medical students	n = 67	25.7	21.31	Three hours of classroom material, a visit to a OIA, and a visit to a healthy adults	na

IG: Intervention Group; CG: Control Group; G: Group; GG: Game group; GNS: Group of neurosciences; GOCA: Group of older community adults; GOIA: Group of older institutionalized adults; GYP: group of young people; OA: Older adults; P-P I: Pre-post intervention.

3.5. Analysis of publication bias

The funnel plot analysis of the main outcome indicated that publication bias could have had an influence upon the meta-analysis, though it must be taken into account that there were studies with a small sample size (n), and the measures were moreover of a continuous and non-dichotomous nature (Fig. S1.3; Supplementary file 1).

3.6. Secondary outcome: change in level of knowledge

An analysis was made of the increase in knowledge in those studies that had examined this parameter based on an instrument used to the effect (Table 2). Regarding the improvement of knowledge about older people, the pooled data evidenced a difference between intervention and control (standardized mean difference = 0.24; 95CI: 0.07–0.4; $p = 0.005$ [heterogeneity: $\text{Tau}^2 = 0.00$; $\text{Chi}^2 = 1.15$, $\text{df} = 4$; $p = 0.89$ $I^2 = 0\%$; $k = 4$; $n = 582$). (Fig. S1.4; Supplementary file 1). The funnel plots indicated that publication bias could have no influence upon the meta-analysis (Fig. S1.5; Supplementary file 1).

4. Discussion

Ageism constitutes discrimination toward older persons mainly due to stereotype-based prejudices. It has been estimated that one out of every two people worldwide has discriminatory attitudes toward older persons. This in turn results in poorer physical and mental health and quality of life among the older population, with an increase in the associated economic costs. The WHO has called for urgent measures to reduce negative attitudes toward the older. The experience gained from the COVID-19 pandemic has illustrated how widespread ageism has become, with the stereotyping of the older and the use of patient age as the sole criterion for facilitating access to medical care, to life-saving therapies and physical isolation measures (World Health Organization, 2021).

University students who after graduating will carry out their professional activities in contact with older people are those who will implement the future health policies referred to health promotion and prevention measures, and will be in charge of assessing, caring for and treating the older population. Although studies have been made of interventions that can improve attitudes toward older persons, their global effectiveness has not been analyzed to date (Burnes et al., 2019; Ross and Williams, 2015; Samra et al., 2017, 2013). To fill this knowledge gap, we completed what to our knowledge constitutes the first meta-analysis of interventions designed to reduce ageism toward older people in undergraduate students.

The present systematic review included 35 pre-post intervention studies, and the meta-analysis identified 14 out of 18 RCTs that assessed the effectiveness of interventions on attitudes toward older people

among undergraduate students. These interventions were classified according to their typology, based on the increase in theoretical knowledge, empathy, the acquisition of practical skills, or a combination of these. All of the studies included in the meta-analysis compared different types of interventions versus control groups. The meta-analyses of the included studies found the overall effects of the interventions to be statistically significant, though empathy-based interventions had the greatest effect.

Empathy comprises two distinct aspects: the ability to comprehend and see the world from the perspectives of others (cognitive empathy), and the ability to connect to the experiences or feelings of others (affective empathy). As can be seen from our findings, interventions based on empathy (Samra et al., 2017, 2013), such as simulations and games (Chen et al., 2015; Demirtas and Basak, 2020; Gholamzadeh et al., 2018; Heliker et al., 1993; Henry et al., 2007; Sari et al., 2020), and contact with older persons (Andreoletti and Howard, 2018; Ng et al., 2020; Ross et al., 2018a, 2018b; Walsh et al., 2008), reduce ageism. Through simulations, students may personally perceive the physical limitations associated with aging. Following these experiences, they may be able to verbalize feelings of defenselessness and resignation toward having to be dependent upon others. They likewise may perceive isolation, feelings of loneliness and depression, and helplessness. In addition to these aspects, consideration is also required of the negative impact produced by the change in body image associated with aging. Following such activities, students will be better able to understand the need for care, help and communication on the part of older people (Sari et al., 2020). Inter-generational contact also appears to be a good tool (Dorfman et al., 2003; Heliker et al., 1993), since by having time to talk about issues that are of interest to the older in a friendly environment, students may be able to get to know and understand them better outside any aggressive environment such as the healthcare setting.

Interventions based on knowledge also increase positive attitudes toward older people, though with a moderate general effect. Although it is believed that increased knowledge has a direct positive impact upon attitudes (Samra et al., 2013), this does not seem to be so clear on the basis of the findings of the present systematic review. It is true that most studies do not contemplate the full program of contents – a fact that may bias these results. Some authors have investigated improvements in attitude using knowledge assessment scales, though it must be underscored that this strategy only affords an indirect measure of attitudes and should not be used as the only instrument, since if knowledge is increased, it is very likely that positive attitudes will also be increased (Samra et al., 2017). The key element in interventions of this kind therefore could be the need to analyze whether such knowledge has truly been acquired, and subsequently analyze attitude.

On the other hand, there are interventions based on the acquisition of clinical skills in healthcare settings, with the participation of practice instructors. Interaction with these professionals also interferes in

Table 2
Characteristics, type of intervention and scores of RCT studies.

Study (author and year)	Country	Population type	Intervention Group, N; mean age (SD-range)/range/percentage of groups of age and % of women	Control Group N, mean age (SD-range)/range/percentage of groups of age and % of women	Type of intervention	Instrument (validity or and reliability in the sample)	Pre M (SD)	Post M (SD)	Intervention description	Duration	Effect size/ significance
Bernard et al., 2003	USA	Medical students	n = 117; 23.6/35.3%	n = 108; 24.9/40.9%	Knowledge + empathy	ASD	CG: 3.42 IG: 3.45	CG: 3.59 IG: 3.85	The Senior mentor program matches the medical student with a mentor over 65 years of age, with whom he/she has to meet at least once per semester, and discussed their history, functionality and risk of geriatric syndromes, as well as expectations.	2 years	na/p = 0.002
Can Gür and Yilmaz, 2020	Turkey	Nursing students	n = 61; total sample: 21.08 (2.18)/69.9%	n = 62; total sample	Empathy	JSSENS ($\alpha = 0.82$)	CG: 94.09 (15.54) IG: 95.11 (13.94)	CG: 86.25 (16.38) IG: 101.14 (15.19)	The IG received mindfulness-based empathy training for 8 weeks, 2 days a week.	8 weeks	d = 0.042/p = 0.024
Demirtas and Basak, 2020	Turkey	Nursing students	n = 59; 20.42 (3.48)/88.1%	n = 60; 20.96 (3.01)/86.7%	Empathy	KAOP ($\alpha = 0.75$)	CG: 90.55 (7.32) IG: 88.45 (7.88)	CG: 93.26 (6.66) IG: 114.33 (5.85)	All students receive a theoretical training course (3 h) related to aging. The IG performs simulation with activities of daily living typical of the older and after debriefing, they complete scales again.	5 weeks	na/p = 0.01
Denton et al., 2009	USA	Medical students	n = 16; 29.4 (4.0)/25%	n = 17; 27.8 (4.2)/41.2%	Knowledge + empathy	A survey	CG: 84.6 (0.16) IG: 81.8 (0.16)	CG: reported 0.5% increase IG: reported 9.8% increase	The intervention group trained via CD room and internet in the development of these objectives. The students made the appointment with their patients and the student each completed their work, which replaced one of the regular compulsory internships.	6 weeks	d = 0.78/p = 0.04
Diachun et al., 2010	Canada	Medical students	n = 108; 24.35 (1.40)/ 57.8%	n = 74; 24.85 (2.62)/58.3%	Knowledge + clinical skills	UCLA GAS	CG: 3.69 (0.43) IG: 3.72 (0.42)	CG: 3.46 (0.55) IG: 3.58 (0.44)	The students completed a practical rotation. IG a clerkship year that did (elder care group) or did not contain a rotation focused on geriatrics. The study was completed with two consecutive cohorts of students.	2 weeks	UCLA GAS ns
Dorfman et al., 2003	USA	Health and social sciences students	n = 13	n = 36	Knowledge + empathy	UCLA Knowledge	CG: 6.75 (1.65) IG: 6.16 (1.62)	CG: 7.49 (1.96) IG: 8.09 (1.69)	The study was conducted using the Service Learning experience methodology. Each student was paired with an older person with whom they carried out different activities and collected their reflections and learning in a reflective diary on a weekly basis.	6 months	UCLA Knowledge p = 0.004 na/ns Intragroup CI p < 0.001
Even-Zohar and Werner, 2020	Israel	Health and social sciences students	n = 186; total sample: 22.3 (3.02)/ 90.3%		Knowledge + empathy	FLTCA	CG: 2.91 (0.36) IG: 2.88 (0.29)	CG: 3.11 (0.38) IG: 3.32 (0.29)	The research is divided into 3 stages. In the first, the proposal was presented to the large group In time 2, the experimental group is given 2 h of training on the care of the elderly, in two ways, a simulation and a viewing of a short film. In the third period, both groups, control and experimental, fill in the scales	2 h	FSA ns Intragroup CI p = 0.01 FAQ1 p = 0.001

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Table 2 (continued)

Study (author and year)	Country	Population type	Intervention Group, N; mean age (SD-range)/range/percentage of groups of age and % of women	Control Group N, mean age (SD-range)/range/percentage of groups of age and % of women	Type of intervention	Instrument (validity or and reliability in the sample)	Pre M (SD)	Post M (SD)	Intervention description	Duration	Effect size/ significance
Fitzgerald et al., 2006	USA	Postgraduate and PhD students of health and social sciences students	n = 33/73%	n = 18/88%	Knowledge + empathy	UCLA GAS	CG: not measured IG: not measured	CG: 4.1 (0.4) IG: 4.1 (0.4)	again, and the experimental group also answers 8 questions posed by the researchers. The IG, completed the CITE training program. Collaborative Interprofessional Team Education (CITE). It includes theoretical training on aging, interactive seminars, skills, planning of interdisciplinary care plans for the older and final reflection.	4 weeks	d = 0.11/ns
						FAQ	CG: na IG: na	CG: 43.3 (18.6) IG: 48.6 (12.9)	The IG is provided with 8 h of empathy training.		d = 0.35/ns
	Turkey	Nursing students	n = 32; 22.7 (1.62)/58.1%	n = 31; 53.1%	Empathy	JSENS	CG: 78.6 (10.1) IG: 77.8 (10.7)	CG: 77.5 (11.3) IG: 85.2 (8.9)	The intervention consisted of an informal interaction between an older adult and a health sciences student for 2 h on 6 occasions (6 h in total). In the sessions, each pair talked and relaxed.	2 days	na/p = 0.004
	USA	Health sciences	n = 11	n = 17	Empathy	KAOP	CG: not measured IG: not measured	CG: not measured IG: 2498% increase	The objectives of the activity-game were explained in 30 min. The participants were provided with accessories for the aging simulation, and each of them visited the stations (places frequented by the elderly in their daily lives), ending with a personal written reflection. The control group received a session with true-false questions to generate discussion. Finally, they also wrote a reflection similar to the participants in the intervention group.	3 weeks	na/p = 0.019
	USA	Health sciences	n = 51; total sample: 84% ≤25/84%	n = 47	Empathy	ASD (α = 0.91) Empathy questions adapted from Maxwell and Sullivan survey	CG: 75.68 (17.7) IG: 74.08 (19.58)	CG: 69.62 (15.59) IG: 68.59 (15.16)	The students completed a set of questionnaires followed by 4 weekly 90-min group sessions, and a post testing session. The training program consisted of four sessions: the psychology of aging, sensory-loss activities associated with normal development in aging, sociodemographic of aging and social interaction skills training with older adults.	1 session; 75–90 min	na/ns
Intrieri et al., 1993	USA	Medical students	n = 45; 26.6/20%	n = 51; 25.7/29.41%	Knowledge + empathy	ASD	CG: 123.9 (18.64) IG: 128.88 (25.07)	CG: 124.07 (18.0) IG: 118.08 (21.32)	Students were assigned a works group with patient old or young. Direct interactions with a client at least 3 times to help change dietary behavior. Finally, after completing the guided experiential assignment, all students wrote responses to 3 reflective items	6 weeks	na/p = 0.003
Lee et al., 2008	USA	Dietetics and/or nutritional sciences students	n = 52; 21.5 (4.0)/42%	n = 48; 20.4 (1.8)/46%	Knowledge + clinical skills	WOAI	CG: 3.49 (0.39) IG: 3.45 (0.35)	CG: 3.45 (0.44) IG: 3.58 (0.42)		3 weeks	na/p < 0.05
						FAQ	CG: 7.3 (2.0) IG: 6.5 (1.8)	CG: 7.1 (2.2) IG: 7.3 (2.1)			na/p < 0.05

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Table 2 (continued)

Study (author and year)	Country	Population type	Intervention Group, N; mean age (SD-range)/range/percentage of groups of age and % of women	Control Group N, mean age (SD-range)/range/percentage of groups of age and % of women	Type of intervention	Instrument (validity or and reliability in the sample)	Pre M (SD)	Post M (SD)	Intervention description	Duration	Effect size/ significance
Leung et al., 2012	China	Health sciences	n = 48; 20.42 (2.20)/ 85.42%	n = 55; 21 (2.19)/ 80%	Knowledge + empathy	KAOP FAQ	CG: 43.72 (5.84) IG: 45.04 (6.48) CG: 42.85 (4.42) IG: 41.48 (3.01)	CG: 47.18 (8.97) IG: 45.83 (20.88) CG: 42.38 (6.6) IG: 44.91 (3.66)	The IG, have a Half-day introductory workshop about older people, 10-week interaction period (1–2 h/week) and half-day intergenerational sharing session. The CG were paired up in groups of two and participated in a 10-week self-directed online learning program on aging.	10 weeks	d = 0.12/p = 0.44 na/p = 0.02
Lindberg and Sullivan, 1996	USA	Medical students	n = 44; 30 (4.4)/ 25%	n = 24; 29.1 (3.5)/42%	Clinical skills + knowledge	MMSAS	CG: 4.04 (0.48) IG: 3.85 (0.55)	CG: 4.06 (0.65) IG: 4.08 (0.48)	The study was developed with tree groups. G1 Experimental with clinical geriatrics rotation with attending Geriatrician; G2 with Experimental same rotation without attending geriatrician and G3 the control group. (G1 and G3 for meta-analysis)	4 weeks	na/na
Lu et al., 2010	USA	Medical students	n = 46; total sample: 23 (1.97)/ 54.7%	n = 91	Knowledge + empathy	ASD ($\alpha = 0.92$) FAQ ($\alpha = 0.86$)	CG: 118.70 (19.26) IG: 116.17 (18.01) CG: 12.31 (3.75) IG: 12.46 (3.28) CG: 34.2 IG: 35.6	CG: 115.07 (18.18) IG: 107.93 (23.26) CG: 12.48 (3.45) IG: 13.09 (3.12) CG: 33.1 IG: 37.2	Training program in geriatrics. The intervention group also participates in the post-test and voluntarily carries out a qualitative focus group.	2 years	na/p = 0.12 na/na
Pacala et al., 1995	USA	Medical students	n = 39/≤25 26%/ 59%	n = 16/≤25 27%/25%	Knowledge + empathy	Pilot instrument using MMSAS ASD ^a	CG: 34.2 IG: 35.6	CG: 33.1 IG: 37.2	The intervention consisted of Aging game exercise. During 6 weeks rotation in ambulatory medicine of 4° degree, and consisted of 30 min introduction, 90 min role playing, simulation (students assuming the roles of elderly adults with loss simulation) in 3 stations. After the role playing, 30 min discussing.	3 h over 5 months	na/ns Intragroup CI p < 0.001
Shue et al., 2005	USA	Medical students	n = 72	n = 89	Knowledge + empathy	MSAS ($\alpha = 0.87/ 0.835$)	CG: not measured IG: 62.02 (8.19)	CG: 64.03 (5.71) IG: 66.5 (5.67)	The intervention group receives a “mentors on aging program” (theory class on aging and meetings with mentors) and the control group are the students from the previous year who did not receive this program. At least 14 occasions, mentor and mentee meet for at least 1 h and work together on aspects of aging. The mentees meet and carry out activities in small groups, which they then discuss with their mentors.	1 year	na/p = 0.001

A survey: Attitudinal survey adapted from Flaherty; ADAS: Age Discrimination Attitude Scale; ASD: Aging semantic differential; CG: control group; FAQ1: facts on aging quiz of Palmore; FLTCA: Foundation for Long term Care Service Learning Project Pillemer & Albright Adapted; FSA: Fraboni's Scale of Ageism; JSENS: Jefferson Empathy Scale; IG: intervention group; KAOP: Kogan's Scale of Attitudes Toward Old People; M ASD: Modified Aging semantic differential; MMSAS: Modified Maxwell-Sullivan Attitude Survey; MSAS: The Maxwell-Sullivan Attitude Survey; na: not available; PNAS, positive negative attitudes scale; ROPE: Relating to older people; UCLA GAS: UCLA Geriatric Attitudinal Scale; UCLA GKT: UCLA Geriatric Knowledge Test; WOAI: Wall-Oyer Aging Inventory.

^a Pilot instrument carried out by the research team, scale not validated.

Table 3

Measurement instruments, type of intervention and results of pre-post intervention studies.

Author and year	Instrument	Validity-reliability in the sample	Groups	Type of intervention	Pre mean (SD)	Post mean (SD)	Percentage change adjusted to range of scale score	Direction attitude change
Alford et al., 2001	Pilot instrument ^a	na	G1	Knowledge + empathy	4.14 (0.77)	4.34 (0.76)	4.00**	+
Andreoletti and Howard, 2018	FSA	$\alpha = 0.74$, -0.62–0.79 for each subscale	G1	Empathy	62.56 (11.69)	60.2 (10.53)	-4.21*	+
Baum and Nelson, 2007	MSAS	na	G1	Knowledge + empathy	3.6 (3.6–3.7)-95% CI	3.7 (3.7–3.8) -95%CI	2.00***	+
Carmel et al., 1992	Pilot instrument ^a	na	GM	Knowledge + empathy	1.2 (0.49)	1.44 (0.84)	na	+
			GN	Knowledge	1.5 (0.79)	1.17 (0.40)	na	+
			GSW	Knowledge	1.41 (0.90)	1.34 (0.44)	-4.96	+
Chen et al., 2015	KAOP	na	G1	Empathy	126.37 (9.21)	153.33 (9.65)	12.87***	+
Deary et al., 1993	Pilot instrument ^a	na	G1	Clinical skills	2.4	2.02	9.5*	+
Duke et al., 2009	R-UCLA GAS	na	G1	Empathy	16.11 (3.98)	14.24 (5.17)	-4.25**	+
Eskildsen and Flacker, 2009	UCLA GAS	na	G1	Empathy	3.7 (0.40)	3.8 (0.40)	2.00***	+
Fields et al., 1992	ASD	na	G1	Knowledge + clinical skills	130.5 (19.20)	126.6 (18.70)	-2.03	+
Goeldlin et al., 2014	UCLA GAS	na	G1	Clinical skills	49 (44–53) IQ range	51 (47–54) IQ range	3.57	+
Gonzales et al., 2010a	ASD (Polizzi)	na	CI	Empathy	75.01 (12.94)	74.02 (1.50)	6.76	+
Haight et al., 1994 (1)	KAOP	na	CG	Knowledge	78.05 (13.59)	64.27 (1.53)	9.56	+
Haight et al., 1994 (2)	ASD	na	G1	Knowledge	na	na	8.12***	+
Henry et al., 2007	ASD	α pre = 0.94 α post = 0.96	G1	Empathy	78.84 (18.2)	82.61 (20.70)	1.96***	+
Hobbs et al., 2006 ^b	UCLA GAS	na	G1	Clinical skills	73.8	73.5	-0.40	-
Hughes et al., 2008	UCLA GAS	na	G1	Knowledge + clinical skills	3.86 (0.36)	3.91 (0.45)	1.00	+
Hwang et al., 2013	R KAOP	$\alpha = 0.86$	GVH	Knowledge + empathy	101.39 (1.81)	98.97 (1.40)	-2.20***	-
			GNH	Knowledge + empathy	102.67 (1.63)	100.95 (1.70)	-1.56***	-
			GAL	Knowledge + empathy	107.25 (1.73)	113.58 (1.69)	5.9***	+
Jeste et al., 2018	COCOA	na	G1	Knowledge	153.6 (17.70)	159.1 (19.60)	2.51***	+
Lee et al., 2005	UCLA GAS	na	GF	Clinical skills	4.07 (0.39)	4 (0.49)	-14.00***	-
			GR	Knowledge + empathy	3.51 (0.43)	3.55 (0.41)	0.8***	+
Lorraine et al., 1998	ASD	na	G1	Empathy	171.56	148.69	-11.91***	+
Ng et al., 2020	KAOP	$\alpha = 0.88$ (positive $\alpha = 0.75$ and negative $\alpha = 0.95$)	G1	Empathy	133 (130.10–135.90) 95%CI	145.7 (143.30–148.10) 95%CI	7.53***	+
Ragan and Bowen, 2001	ASD	na	IG	Knowledge	96.55 (16.52)	116.27 (22.38)	10.27	+
			CLG	Knowledge + empathy	101.15 (20.87)	120.42 (18.72)	10.04*	+
			RIG	Knowledge	100.73 (22.63)	171.52 (19.91)	70.27*	+
Robinson and Rosher, 2001	ASD	na	G1	Empathy	124.35 (68–174) 95%CI	116.14 (66–173) 95%CI	-4.28**	+
Ross and Williams, 2015	ASD	na	G1	Empathy	107.3	102.5	-2.50	+
Ryan et al., 2007	KAOP	$R = 0.66$ –0.83 α pre = 0.8 α post = 0.81	G1	Knowledge + empathy	126.4 (9.99)	128 (9.54)	0.62	+
Sarabia-Cobo and Castanedo Pfeiffer, 2015	CENVE ^c	na	G1	Knowledge	65.96	16.73	48.62**	+
Sari et al., 2020	JSENS-HPS	na	na	Empathy	111.41 (11.41)	117.39 (10.78)	4.98**	+
Silver et al., 2016 (1) ^d	ERA 12	$\alpha = 0.35$ –0.52	G1	Knowledge + empathy	45.86 (9.05)	53.49 (10.47)	7.63**	+
Silver et al., 2016 (2)	ERA MH	$\alpha = -0.6$ –0.33	G1	Knowledge + empathy	48.86 (12.02)	65.36 (16.19)	16.50**	+
Stewart et al., 2007	M ASD	na	G1	Knowledge	2.8 (0.50)	2.8 (0.40)	0.01**	=
			G2	Knowledge	2.8 (0.40)	2.9 (0.90)	2.5**	+
			G3	Knowledge	2.9 (0.4)	2.7 (0.3)	-3.44**	+
			G4	Knowledge	2.9 (0.4)	2.7 (0.4)	-6.89*	+
Van De Pol et al., 2014	ASD	na	G1	Knowledge	84 (11)	75 (15)	-3.65**	+

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Table 3 (continued)

Author and year	Instrument	Validity-reliability in the sample	Groups	Type of intervention	Pre mean (SD)	Post mean (SD)	Percentage change adjusted to range of scale score	Direction attitude change
Walsh et al., 2008	RKAOP	$R = 0.79-0.46$ α pre = 0.8 α post = 0.85	GCBF GFV	Empathy Empathy	68.45 80.73	na 63.37	na 15.78***	= +
Warren et al., 1983	M Kafer and M MSAS	na	G1	Knowledge + clinical skills	72.6 (7.40)	77.3 (7.1)	6.27***	+
Westmoreland et al., 2009 ^d	UCLA GAS	na	GPre GPost	— Knowledge + empathy	na na	na na	na na	na na
Wilkinson et al., 2002	ASD (Polizzi)	na	GOCA+GOIA GYP	Empathy Empathy	121.7 118.2	108.4 155.7	-9.24*** -1.79	+ +
Williams et al., 2007	ASD	na	G1	Knowledge + clinical skills	107.5	104.33	-1.65	+
Wilson and Glamser, 1982	ASD	na	G1	Knowledge + empathy	130.7 (18.78)	120.16 (20.98)	-5.49***	+

ASD (Polizzi): Aging semantic differential revised by Polizzi; ASD: Aging semantic differential; CENVE: The Questionnaire of Negative Stereotypes about Aging; CLG: Campus Life Group; COCOA: Carolina Opinions on Care of Older Adults; ERA 12: Expectations Regarding Aging; ERA-MH: Expectations Regarding Aging Mental Health; FSA: Fraboni's Scale of Ageism; G1 and G2 Received half new educational medical program in two different academic years; G3 and G4 Received complete new educational medical program in two different academic years; GAL Group of assisted living facilities; GCBF: Group of Creative bonding intervention; GF Group of fellows; GR Group of residents; GFV Group of Friendly Visits; GG Game Group; GM Group of medical students; GN Group of Nursing students; GNH Group of Nursing Homes; GNS Group of Neurosciences; GOCA Group of Older Community Adults; GOIA Group of Older Institutionalized Adults; GPost Group post intervention; GPre Group pre intervention; GSW Group of Social Worker Students; GVH Group of Veteran's Home; GYP group of young people; IG: Information Group; IQ range: interquartile range; KAOP: Kogan's Scale of Attitudes Toward Old People; M ASD: Modified Aging semantic differential; M Kafer: Modified Kafer; M: MSAS: The Maxwell-Sullivan Attitude Survey; na: not available; OA: Older Adults; OCA: Older Community Adults; OHA: Older Hospitalized Adults; OIA: Older Institutionalized Adults; R: reliability; R- UCLA GAS: Revised Geriatric Attitudinal Scale; RIG: Reinforcement information group; RKAOP: Revised Kogan's Scale of Attitudes Toward Old People; UCLA GAS: UCLA Geriatric Attitudinal Scale; MSAS: Modified Maxwell Sullivan; α : alpha de Cronbach; 95CI: Confident interval of 95%.

^a Pilot instrument carried out by the research team, scale not validated.

^b Scores refer to percentage of positive answers.

^c Scores refer to percentage of negative answers.

^d Differences in six items with more positive after intervention; (1) and (2): the same study assessed attitudes using two instruments.

* Significant at <0.05.

** Significant at <0.01.

*** Significant at <0.001.

interventions of this kind, since more positive attitudes on the part of the healthcare professional will increase quality of care, while negative attitudes could even delay or impede care and the adequate treatment of older persons (Eymard and Douglas, 2012). The existence of ageism in healthcare settings therefore makes it necessary to target interventions not only to students but also to administrators, physicians, nurses, personal caregivers and other associated healthcare professionals (Tullo et al., 2010). On the other hand, interaction with the student is not only limited to practice instructors but also includes the lecturers that teach the subjects in the University. In the same way as in the case of the practice instructors, their attitudes and knowledge can have a direct influence upon the students (Eymard and Douglas, 2012; Hogan et al., 2014; Negrin et al., 2020), since the way in which the lecturer explains the concept and process of aging may have a positive or negative impact upon the students. Experiences as volunteers in friendly environments in centers for the older, or with older relatives, allow students to know the reality of the environment in while older people live, and increase positive attitudes (Millns Sizer et al., 2016). In contrast, practices for the acquisition of clinical skills carried out in early academic courses without adequate knowledge about older people, or the existence of shortcomings in terms of practice instructor attitudes and knowledge, may lead to real situations in which the students do not know how to behave and resolve problems. This would cause practices, which are generally a positive experience, to become totally negative in this regard, with a possible adverse impact upon the attitudes of the future professionals (Karikari et al., 2020). Thus, in the acquisition of clinical skills, the attitudes and circumstances of the professionals in the healthcare center also play a direct role.

Despite the findings of our study, the comparisons of the different studies were complicated by the presence of a number of factors such as the heterogeneity of the interventions (Burnes et al., 2019; Ross et al., 2018a, 2018b; Samra et al., 2013, 2017), their duration (Tullo et al.,

2010), and the cultural aspects inherent to the study sample and students – with the aging process being viewed differently depending on the cultural setting involved (Seah et al., 2019).

In addition to the different interventions, consideration is also required of those factors which in one way or other influence attitudes among students. In this regard, the female sex, an age of over 25 years, married status, prior knowledge of gerontology, full-time dedication to study, and a history of contact with older people have a positive impact upon student attitudes (Hobbs et al., 2006; Karikari et al., 2020; Samra et al., 2017; Veronek et al., 2020). In contrast, the fact of having previously worked in the geriatric setting has a negative impact upon attitudes (Veronek et al., 2020).

In view of the reported lack of well-designed studies (Karikari et al., 2020; Koskinen et al., 2015; Lee, 2015; Samra et al., 2017, 2013) allowing analysis of the effectiveness of the different interventions in students, and considering the results obtained, we can propose a series of characteristics that should be incorporated to the design of intervention studies seeking to improve attitudes toward the older:

1. Randomized clinical trials are advised, since it is necessary to include a control group with which to compare the proposed intervention and thus be able to draw conclusions.
2. The need to use validated scales for assessing attitudes toward older persons according to the study population involved. It is not advisable to modify the scales or use indirect measures such as knowledge or attitude toward working with older people.
3. The validity and reliability of the scale applied to the study sample should be analyzed.
4. The intervention should be described in maximum detail. In the case of knowledge-based interventions, the program should be described in full.

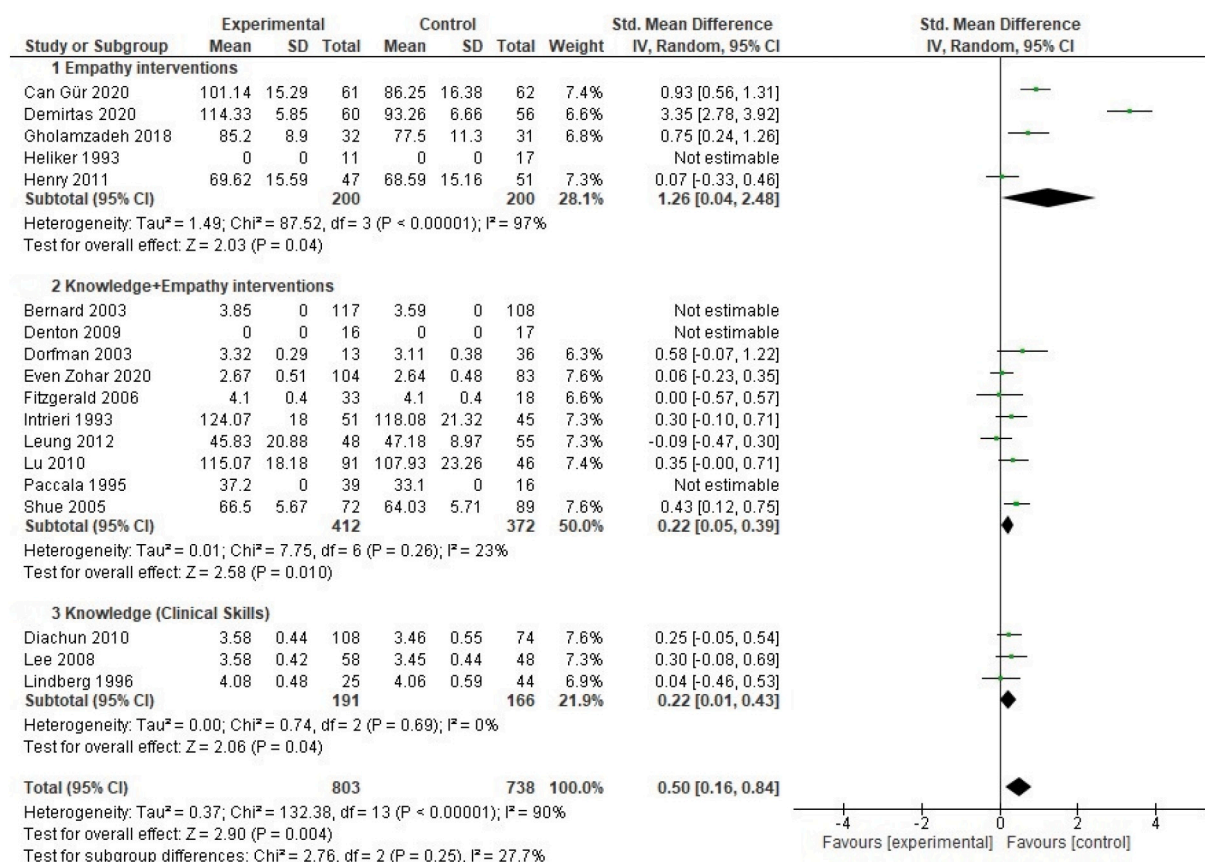


Fig. 2. Meta-analysis of standard mean differences for continuous variables (random effects RE model) in attitudes toward older people scales.

- The intervention should be classified as being based on knowledge, clinical skills, empathy or their combination.
- It is important to determine whether there has been an acquisition of knowledge, clinical skills and empathy, in addition to the analysis of attitudes following the intervention.
- Attitudes should be analyzed according to the following variables related to the students: sex, age, marital status, previous knowledge (courses in geriatrics and gerontology), previous experiences with relatives, and previous work with older people.
- Attitudes should be analyzed according to variables related to both the lecturers and practice instructors (years of experience and training in geriatrics).

4.1. Strengths and limitations

The present systematic review and meta-analysis is the first to include randomized clinical trials in University students and to analyze pre-post intervention studies seeking to improve attitudes toward older people. In order to be able to compare the pre-post intervention studies, we performed a calculation of percentage change unifying the measurement units. In addition, we only contemplated studies that analyzed attitude based on direct measurement scales. From the results obtained, future lines of research can be proposed, allowing the comparison of outcomes with the smallest bias possible.

The limitations of the study arise from the need to perform a systematic review without time restrictions, in view of the lack of previous studies. This produced a very wide range of years that increased the variability of contents and of the University curricular programs. Furthermore, there were studies that published incomplete results – a fact that prevented us from performing a full meta-analysis. Most of the studies included in the systematic review were pre-post studies lacking a

control group, and therefore could not be included in the meta-analysis. Some authors in turn modified the measurement scale and/or generated their own measurement instruments, without due confirmation of sample validity and reliability. Categorization of the types of interventions was determined from the information reflected in the methodology of the studies, since not all the descriptions of the interventions were clear in relation to the intended objectives. For example, after addressing clinical skills, work possibly could also have been done on empathy. We were unable to establish an analysis according to sex and age due to the lack of data, and this could also have biased the results obtained. In turn, publication bias could have influenced the meta-analysis in relation to the main outcome. Lastly, it must be noted that abundant literature is available in this regard, and possibly not all the existing studies were included.

5. Conclusions

This is the first meta-analysis to examine the efficacy of interventions seeking to improve attitudes toward the older population among university students in the health and social sciences. Despite the abundant existing literature, only 14 studies could be included in this meta-analysis, which found the overall effects of the interventions to be statistically significant, though empathy-based interventions had the greatest effect in terms of improving attitudes toward the older. Comparisons of the different studies were made difficult by the fact that there were few well-designed studies, with the presence of a number of complicating factors such as the heterogeneity of the interventions, their duration, and cultural issues. Well-designed studies that take into account the recommendations made after the analysis of results are much needed in order to improve our knowledge of the effect of different interventions targeted to future professionals in charge of the care of older people.

CRediT authorship contribution statement

Francisco M Martínez-Arnu: Resources, Data curation, Writing-Original draft preparation, Project administration. **Lourdes López-Hernández:** Data curation, Writing- Original draft preparation. **Marta Botella-Navas:** Data curation, Writing- Original draft preparation. **Elena Castellano-Rioja:** Data curation, Writing- Original draft preparation; **Pilar Pérez-Ros:** Conceptualization, Methodology, Resources, Software; Writing- Reviewing and Editing.

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Declaration of competing interest

None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.nedt.2022.105269>.

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